

Value Function Approximation — Rollout Layer

wraps any base policy above; overrides its greedy drop decision via short-horizon Monte-Carlo rollout when a candidate wins $\geq 70\%$ of scenarios

Myopic

$$\begin{aligned}\text{score} &= 1/d_{\text{cur},i} \\ \text{drop} &= 1/\Delta t_i\end{aligned}$$

greedy, no economic weighting

Myopic+

$$\begin{aligned}\text{score} &= p_i/d_{\text{cur},i} \\ \text{drop} &= p_i/\Delta t_i\end{aligned}$$

+ power-weighted urgency

CFA

$$\begin{aligned}\text{score} &= (\tilde{C}_i + s)/d_{\text{cur},i} \\ \text{drop} &= \tilde{C}_i - \frac{c_L}{60} \Delta t_i\end{aligned}$$

+ learned station value $\tilde{C}$

Dynamic Balance

$$\begin{aligned}\text{score} &= (\tilde{C}_i + s)/d_{\text{cur},i}^{2\delta} \\ \text{drop} &= \tilde{C}_i - 2\delta \frac{c_L}{60} \Delta t_i\end{aligned}$$

+ state-dependent balance δ

Shared Zone Selection Layer

K-Means zones, scored centrality-based or value-based (policy-specific station value)